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BAND SAW

Abstract

A band saw includes a housing having a handle configured to be grasped by a user during a cutting operation, a motor supported by the housing, and a drive wheel assembly rotationally driven by the motor. The drive wheel assembly has a drive wheel and a driven wheel. The band saw further includes an opening defined within the housing and located between the drive wheel and the driven wheel, an endless saw blade driven by the drive wheel assembly and configured to pass through the opening to cut a workpiece during the cutting operation, and a transmission assembly. The transmission assembly has an input member for receiving torque from the motor and an output member for transmitting torque to the drive wheel about a drive wheel axis. The transmission assembly includes at least one cycloidal gear stage.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/560,822, filed Mar. 4, 2024, the entire content of which is incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to power tools, and more specifically to band saws.

BACKGROUND OF THE INVENTION

[0003] Handheld band saws are used to cut a variety of workpieces, such as pipes, rods, studs, and blocks. A cordless, compact band saw allows for cutting operations in tight spaces or awkward angles for plumbing, electrical, remodeling, and HVAC applications.

SUMMARY OF THE INVENTION

[0004] The invention provides, in one aspect, a band saw including a housing having a handle configured to be grasped by a user during a cutting operation, a motor supported by the housing, and a drive wheel assembly rotationally driven by the motor. The drive wheel assembly includes a drive wheel and a driven wheel. The housing defines an opening located between the drive wheel and the driven wheel. The drive wheel assembly drives an endless saw blade that is configured to pass through the opening to cut a workpiece during the cutting operation. The band saw further includes a transmission assembly having an input member for receiving torque from the motor and an output member for transmitting torque to the drive wheel about a drive wheel axis. The transmission assembly includes at least one cycloidal gear stage.

[0005] The invention provides, in another aspect, a band saw including a housing having a handle configured to be grasped by a user during a cutting operation and a motor supported by the housing. The motor has an outer rotor and an inner stator at least partially received within and surrounded by the outer rotor. The band saw further includes a drive wheel assembly rotationally driven by the motor. The drive wheel assembly has a drive wheel and a driven wheel. Moreover, the band saw includes an opening defined within the housing and located between the drive wheel and the driven wheel, an endless saw blade driven by the drive wheel assembly and configured to pass through the opening to cut a workpiece during the cutting operation, and a transmission assembly. The transmission assembly has an input member for receiving torque from the motor and an output member for transmitting torque to the drive wheel about a drive wheel axis. The transmission assembly includes at least one cycloidal gear stage.

[0006] The invention provides, in another aspect, a band saw including a housing having a handle configured to be grasped by a user during a cutting operation and a motor supported by the housing. The motor has an outer rotor and an inner stator at least partially received within and surrounded by the outer rotor. The band saw further includes a drive wheel assembly rotationally driven by the motor. The drive wheel assembly has a drive wheel and a driven wheel. Moreover, the band saw includes an opening defined within the housing and located between the drive wheel and the driven wheel, and an endless saw blade driven by the drive wheel assembly. The endless saw blade is configured to pass through the opening to cut a workpiece during the cutting operation.

Description

BRIEF DESCRIPTION OF DRAWING

[0007] FIG. **1** is perspective view of a band saw in accordance with an embodiment of the invention.

[0008] FIG. **2** is an exploded view of a portion of the band saw of FIG. **1**.

[0009] FIG. **3** is a cross-sectional view of the band saw through section line 3-3 in FIG. **1**.

[0010] FIG. **4** is an exploded perspective view of a motor for use with the band saw of FIG. **1**.

[0011] FIG. **5** is a reverse exploded perspective view of the motor of FIG. **4**.

[0012] FIG. **6** is a cross-sectional view of a band saw according to another embodiment.

[0013] Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DETAILED DESCRIPTION

[0014] FIGS. **1-3** illustrate a band saw **10** including a housing **14** supporting a drive mechanism. The drive mechanism includes a motor **18** and a transmission assembly **16**. In the illustrated construction of the band saw **10**, the motor **18** is configured as a brushless DC (BLDC) electric motor, and the band saw **10** includes a removable, rechargeable battery pack **26** for supplying electrical current to the motor **18**. The drive mechanism further includes a drive wheel assembly **30**, which is driven by the motor **18** and transmitted torque via the transmission assembly **16**. The motor **18**, the transmission assembly **16**, and the drive wheel assembly **30** are supported by the housing **14**.

[0015] With reference to FIG. **1**, the housing **14** includes a main or first handle **38** and an auxiliary or second handle **40**. The first handle **38** is provided with a trigger **42** to selectively supply power to the band saw **10**. The trigger **42** is disposed adjacent a gripping portion **44** of the first handle **38** where a user grasps the band saw **10** during a cutting operation. In the illustrated construction of the band saw **10**, the battery pack **26** is supported by the housing **14** between the first handle **38** and the second handle **40**. The trigger **42** is operable to control operation of the motor **18**. Specifically, the battery pack **26** selectively supplies electrical current to the motor **18** when the trigger **42** is depressed.

[0016] The housing **14** of the band saw **10** also includes a deck **46** and a guard **50** coupled to the deck **46**. The deck **46** and the guard **50** form a cavity (not shown) in which the drive wheel assembly **30** and an endless band saw blade **32** are at least partially positioned. The deck **46** defines a cut zone **58**. The deck **46** and the guard **50** substantially cover the band saw blade **32** when the blade **32** is in a shielded position (i.e., when the blade **32** is not within the cut zone **58**). However, the deck **46** and the guard **50** do not cover the band saw blade **32** when the blade **32** is in an exposed position (i.e., when the blade **32** passes through the cut zone **58**). In the exposed position, the blade **32** is fully exposed and unobstructed by the guard **50**, allowing workpieces to be cut when entering the cut zone **58**. The band saw **10** further includes a shoe **66** coupled to the housing **14** and having a support surface **74** against which a work piece is abutted during a cutting operation.

[0017] The housing **14** includes a motor housing portion **62** extending from the deck **46**. The motor housing portion **62** encloses the motor **18**. The motor housing portion **62** extends a height dimension H (FIG. **1**) from the deck **46**. In the illustrated embodiment, the height dimension H is between 20 mm and 30 mm. More particularly, the height dimension H is 25.5 mm. As explained in further detail below, the length of the transmission assembly **16** is less than that of a traditional band saw in which a multi-stage planetary transmission is positioned between a drive wheel and the electric motor, which requires a motor housing height much greater than the height dimension H in

the band saw **10**.

[0018] With continued reference to FIG. **1**, the housing **14** of the band saw **10** includes an electronics housing portion **68** coupled to the deck **46** and disposed above the cut zone **58**. The electronics housing portion **68** is also disposed between the first handle **38** and the second handle **40**. Electronic components (e.g., a power and/or control printed circuit board; not shown) used to operate the band saw **10** are housed within the electronics housing portion **68**. A battery receptacle (not shown) is formed on the electronics housing portion **68** and configured to receive the battery pack **26**. As such, the electronic components within the electronics housing portion **68** electrically connect the battery pack **26** to the motor **18**.

[0019] With reference to FIGS. **2** and **3**, the drive wheel assembly **30** includes a drive wheel **70** that is driven by the motor **18** and a driven wheel (not shown) that is driven by the drive wheel **70** via the saw blade **32**. Portions of the drive wheel **70** and the driven wheel are covered by the deck **46** and the guard **50**. The cut zone **58** is located between the drive wheel **70** and the driven wheel, and coincides with an opening **64** in the housing **14** and/or deck **46** through which the saw blade **32** passes. The drive wheel **70** rotates about a drive wheel axis **78** that is coaxial with a rotational axis or a first axis **84** of an input member or input shaft **80** of the transmission assembly **16** which, in turn, receives torque from the motor **18**. As is described in detail below, the transmission assembly **16** receives torque from the motor **18** and provides torque through an output member to the drive wheel **70**, causing the drive wheel **70** to rotate about the drive wheel axis **78**. A tire **88** is coupled to the outer peripheral surface of each of the drive wheel **70** and the driven wheel. The tire **88** grips the saw blade **32** during a cutting operation to reduce or prevent slippage between the drive wheel **70** and the driven wheel, respectively, with the saw blade **32**. The saw blade **32** is driven by the drive wheel assembly **30** and passes through the cut zone **58** during the cutting operation.

[0020] With continued reference to FIGS. **2** and **3**, the transmission assembly **16** includes a single stage cycloidal gear set **20**. In other embodiments, the transmission assembly **16** may include two or more cycloidal gear stages. The transmission assembly **16** includes a transmission housing **22** coupled to the housing **14** and/or the deck **46**. In the illustrated embodiment, the transmission housing **22** includes a frame **118** and a cover **72** affixed to the frame **118** (e.g., with fasteners). In other embodiments, the frame **118** and cover **72** may be affixed using integral features on the frame **118** and cover **72**, respectively (e.g., snap-fits). The single stage cycloidal gear set **20** is housed in the transmission housing **22**.

[0021] The single stage cycloidal gear set **20** includes a first cam **92** and a first disc **100**. The first cam **92** is coupled to the input shaft **80**, such that the first cam **92** is rotated about the first axis **84** by the input shaft **80**. The first cam **92** drives the first disc **100** into an orbiting motion about the first axis **84**. The first disc **100** defines a second axis **86** through a center of the first disc **100**. The first disc **100** rotates about the second axis **86** and orbits about the first axis **84** in response to rotation of the first cam **92** about the first axis **84**.

[0022] With continued reference to FIGS. **2-3**, the transmission assembly **16** includes a plurality of cylindrical pins **122** having a first end **114** and a second end **116** opposite the first end **114**. The cylindrical pins **122** define a circular cross-section when viewed along the drive wheel axis **78**. The first end **114** of each of the cylindrical pins **122** is anchored in a blind bore **130** of the cover **72**. The second end **116** of each of the cylindrical pins **122** is anchored in a blind bore **132** the frame **118**. As such, the cylindrical pins **122** extend between the cover **72** and the frame **118** and are affixed within the transmission housing **22**. In some embodiments, the cylindrical pins **122** are slip-fit with the blind bores **130**, **132** of the cover **72** and the frame **118**. As the cover **72** and the frame **118** are secured together during assembly, a nominal clamping force is applied to the cylindrical pins **122** to restrict the movement of the cylindrical pins **122** within the transmission housing **22**. In other embodiments, the cylindrical pins **122** may be press-fit into the blind bores **130**, **132** of the cover **72** and/or the frame **118**.

[0023] The use of cylindrical pins **122** reduces manufacturing costs of the transmission assembly

16 when compared to radially inward extending teeth to engage with a portion of the single stage cycloidal gear set **20**. The use of radially inward extending teeth, as known in the prior art, requires additional machining due to the complex geometry of the teeth. The additional machining increases overall time and manufacturing cost of the transmission housing **22**. The cylindrical pins **122** used in the disclosed embodiment may be constructed from a cylindrical bar stock material (e.g., Aluminum, steel, etc.). The cylindrical bar may be cut to length and affixed to the transmission housing **22** using any of the methods described above at a reduced cost.

[0024] With reference to FIG. 2, the drive wheel **70** includes a plurality of drive lugs **108** extending from the drive wheel **70** in a direction parallel with the drive wheel axis **78**. The drive lugs **108** include an outer diameter **136** that form a generally cylindrical cross-section viewed along the drive wheel axis **78**. The drive lugs **108** are coupled to the drive wheel **70** for co-rotation about the drive wheel axis **78**.

[0025] With reference to FIGS. 2-3, the first disc **100** includes a first plurality of apertures **110** to receive the drive lugs **108**. The apertures **110** define an inner diameter **128** that is greater than the outer diameter **136** of the drive lugs **108** to provide clearance for the drive lugs **108** when received in the corresponding apertures **110** to not obstruct the rotational and orbital (i.e., cycloidal) movement of the first disc **100** within the transmission housing **22**. Specifically, the inner diameter **128** of the apertures **110** imparts a rotational force on each of the drive lugs **108**, thereby applying torque to the drive wheel **70** to cause it to rotate. The first disc **100** further includes a first plurality of rounded teeth **102** on an outer circumference of the first disc **100**. The teeth **102** selectively engage the cylindrical pins **122** as the first disc **100** both rotates and orbits within the transmission housing **22**.

[0026] The transmission assembly **16** includes a vibration attenuation assembly **24** to reduce vibration caused by the single stage cycloidal gear set **20**. The vibration attenuation assembly **24** includes a second cam **96** and a second disc **104**. The input shaft **80** provides a torque input from the motor **18** to the second cam **96**, such that the second cam **96** rotates about the first axis **84**. The second disc **104** defines a third axis **90** through a center of the second disc **104**. The second disc **104** rotates about the third axis **90** and orbits about the first axis **84** in response to rotation of the second cam **96** about the first axis **84**. The vibration attenuation assembly **24** is operated 180 degrees out of phase with the single stage cycloidal gear set **20**. Specifically, the first cam **92** is 180 degrees out of phase with the second cam **96** and the first disc **100** is 180 degrees out of phase with the second disc **104**, thereby counteracting unbalanced forces caused by the cycloidal motion of the single stage cycloidal gear stage **20** and attenuating resultant vibration.

[0027] The second disc **104** includes a second plurality of apertures **112** having an inner diameter **140**. The inner diameter **140** of the apertures **112** is greater than the outer diameter **136** of the drive lugs **108** to provide clearance during operation. The drive lugs **108** extend through the apertures **110** and the apertures **112**. The second disc **104** further includes a second plurality of rounded teeth **106** on an outer circumference of the second disc **104**. The teeth **106** selectively engage with the cylindrical pins **122** as the second disc **104** both rotates and orbits within the transmission housing **22**.

[0028] The cylindrical pins **122** include a number of pins N , and the first disc **100** and the second disc **104** include a number of teeth M , equal to one less than the number of pins N (i.e., $M=N-1$). The number of pins N and the number of teeth M are to mesh upon rotation of the input shaft **80**. When the input shaft **80** rotates, a reduction ratio R is governed by the following equation:

$$[00001] R = \frac{M}{N-M}$$

[0029] In the illustrated embodiment shown in FIG. 2, the cylindrical pins **122** includes 45 pins. The first disc **100** and the second disc **104** each include 232 teeth. Therefore, the transmission assembly **16** provides a reduction ratio of 232:1. In other words, the first disc **100** rotates about the drive wheel axis **78** to rotate the drive lugs **108** about 8.18 degrees for every 360 degree rotation of the first cam **92**. In other embodiments, the first disc **100** may provide any other reduction ratio R

as desired.

[0030] In operation, the user depresses the trigger **42** to activate the motor **18** with electrical current from the battery pack **26**. The motor **18** provides torque to the shaft **80**, causing it to rotate about the first axis **84**. With the first cam **92** being coupled for co-rotation with the shaft **80**, the first cam **92** is co-rotated with the shaft **80**. The first disc **100** is driven to orbit about the first axis **84** due to an eccentric offset of the first cam **92** with the shaft **80**. The first disc **100** is also driven to rotate about the second axis **86** defined through the center of the first disc **100**. The cycloidal motion of the single cycloidal gear stage **20** is a result of the first disc **100** rotating about the second axis **86** and orbiting about the first axis **84** simultaneously. The cycloidal motion of the first disc **100** within the transmission housing **22**, with the drive lugs **108** received within the apertures **110** in the first disc **100**, is transmitted to the drive wheel **70** as rotation. The shaft **80** also provides a torque input to the vibration attenuation assembly **24**. With the second cam **96** being coupled for co-rotation with the shaft **80**, the second cam **96** is co-rotated with the shaft **80**. The second disc **104** is driven to orbit about the first axis **84** due to an eccentric offset of the second cam **96** with the shaft **80**. The second disc **104** is also driven to rotate about the third axis **90** defined through the center of the second disc **104**. The cycloidal motion of the vibration attenuation assembly **24** is a result of the second disc **104** rotating about the third axis **90** and orbiting about the first axis **84** simultaneously. In use, the vibration attenuation assembly **24** moves 180 degrees out of phase with the single cycloidal gear stage **20**. Therefore, the vibration attenuation assembly **24** counteracts unbalanced forces caused by the cycloidal motion of the single stage cycloidal gear stage **20** and attenuates resultant vibration.

[0031] FIGS. **4** and **5** illustrate an alternative motor **18'** that may be incorporated into the band saw **10** of FIGS. **1-3**. The motor **18'** is a brushless direct current ("BLDC") electric motor with a stator **200** and a rotor **204** that is rotatable with respect to the stator **200** about an axis **78'**. The BLDC electric motor **18'** is an "outer rotor" BLDC electric motor with the rotor **204** generally surrounding the stator **200**. As such, the rotor **204** is an outer rotor **204** and the stator **200** is an inner stator **200** that is at least partially received within and generally surrounded by the outer rotor **204**.

[0032] The stator **200** includes a stator mount **208**, a stator core assembly **212** fixedly supported by the stator mount **208**, and a plurality of wires or stator windings that define a plurality of coils **218**. The stator core assembly **212** includes a stator core **220** formed, in the illustrated construction, by a stack of laminations, and an insulator **224** (FIG. **5**) molded to the stator core **220**. The stator core **220** includes a plurality of teeth **226** in which the coils **218** are formed about the teeth **226** and insulated therefrom by the insulator **224**. The stator core **220** defines a core central bore **228** that extends longitudinally therethrough. The stator mount **208** includes an elongated stator support portion **232** and a motor support portion **236** located at one longitudinal end of the stator support portion **232**. The stator support portion **232** is tubular in shape and supports the stator core assembly **212**. More specifically, the core central bore **228** of the stator core **220** receives the stator support portion **232**, such that the stator core assembly **212** is rigidly supported about the stator support portion **232**. In some examples, the stator core **220** can receive the stator support portion **232** by press fit or interference fit. In other examples, the stator support portion **232** can be affixed to the stator core **220** by a molding process. The stator mount **208** further defines a mount central bore **240** that extends longitudinally therethrough, including through both the stator support portion **232** and the motor support portion **236**.

[0033] With continued reference to FIGS. **4** and **5**, the rotor **204** includes a central rotor shaft **244**, a rotor frame **248**, a tubular rotor body **252**, and a plurality of permanent magnets **256**. The rotor frame **248** is annular in shape and includes a central portion **260A** that is affixed to the rotor shaft **244**. An outer peripheral portion **260B** of the rotor frame **248** is affixed to the rotor body **252**. As such, the rotor frame **248** affixes the rotor body **252** to the rotor shaft **244** for co-rotation therewith. A plurality of radially and axially extending blades **260C** extend between and connect the central portion **260A** to the outer peripheral portion **260B**, and a plurality of airflow apertures **260D** are

defined between each pair of adjacent blades **260C**. The blades **260C** operate as a fan to generate an airflow that passes through the airflow apertures, thereby cooling the motor **18'**. In some embodiments, the rotor frame **248** can be formed from a metal or metal alloy such as, e.g., zinc or steel. In other embodiments, the rotor frame **248** can be molded from a resin material.

[0034] The rotor body **252** is tubular in shape and defines a central cavity **264**. The permanent magnets **256** are fixedly supported on a radially inner surface of the rotor body **252**. When assembled, the rotor body **252** is located radially outward from the stator **200** and surrounds portions of the stator **200**, including the stator core assembly **212**, the coils, and portions of the stator mount **208**. Portions of the stator **200**, including all or generally most of the stator core assembly **212**, are received into the central cavity **264** of the rotor body **252**.

[0035] The rotor shaft **244** is rotatably supported relative to the stator **200** by two bearings including a first bearing **268** and a second bearing **272**. As such, the rotor **204** (i.e., including the rotor shaft **244**, the rotor frame **248**, the rotor body **252**, and the magnets **256**) rotates relative to the stator **200**. When the motor **18'** is supported within the motor housing portion **62** of the band saw **10**, the rotor **204** is rotatable relative to both the motor housing portion **62** and the stator **200**, whereas the stator **200** does not rotate relative to the motor housing portion **62**. The rotor shaft **244** extends centrally through the stator **200**, both through the core central bore **228** and the mount central bore **240**.

[0036] The motor **18'** further includes an annular printed circuit board assembly (PCBA) **276** affixed to the insulator **224** of the stator **200**, as will be described in greater detail below. The PCBA **276** can include at least one position sensor, such as a Hall effect sensor, operable to detect a position of the permanent magnets **256** of the rotor **204**. In addition, or alternatively, the PCBA **276** can include a plurality of switching circuits (e.g., Field-Effect Transistors (FETs)) operable to electrically commutate the motor **18'**.

[0037] FIG. **6** illustrates a band saw **310** according to another embodiment. The band saw **310** is similar in some aspects to the band saw **10** of FIGS. **1-3** described above, and features of the band saw **310** corresponding with features of the band saw **10** are given like reference numerals plus '**300**.' In addition, the following description focuses primarily on differences between the band saw **310** and the band saw **10**, and it should be understood that features of the band saw **10** may be incorporated into the band saw **310**, and vice versa.

[0038] The band saw **310** includes a housing **314** supporting a drive mechanism. The drive mechanism includes a motor **318** and a transmission assembly (not shown). The transmission assembly may take the form of a multi-stage planetary transmission or a transmission including the single stage cycloidal gear set **20** of FIGS. **2** and **3**. In the illustrated construction of the band saw **310**, the motor **318** is disposed within a motor housing portion **362** of the housing **314** and is configured as an "outer rotor" BLDC electric motor. The motor **318** has a stator **500** and a rotor **504** that is rotatable with respect to the stator **500**. The rotor **504** is an outer rotor **504** and the stator **500** is an inner stator **500** that is at least partially received within and generally surrounded by the outer rotor **504**. The stator **500** is supported by a stator frame **508** that may be coupled to the housing **314** or the motor housing portion **362**. Also, the rotor **504** has a rotor shaft **512** operably coupled to the transmission assembly to rotate a drive wheel **370** during operation of the band saw **310**. The rotor shaft **512** is rotatably supported relative to the stator **500** by two bearings including a first bearing **516** and a second bearing **520**.

[0039] Due to the motor **18'**, **318** being an "outer rotor" BLDC electric motor, the size of the band saw **10**, **310** is reduced. In comparison to conventional band saws that may have an "inner rotor" BLDC motor, the band saw **10**, **310** has at least 15 fewer parts and weighs at least 0.5 lbs. less. In this alternate embodiment, the height dimension H_{is} is about 22 mm. As such, in some embodiments, the motor **18'**, **318** may nest within the drive wheel **70**, **370** to thereby reduce the size of the band saw **10**, **310**. Size reduction of the band saw **10**, **310** gives the user the ability to perform a flush cutting operation by reducing the height of the housing **14**, **314** up to about $\frac{3}{8}$ of an inch compared

to known band saws with inner rotor BLDC motors. Size reduction also increases durability of the band saw **10, 310** by reducing the drop risk of the band saw **10, 310**.

[0040] In addition, the configuration of the motor **18', 318** yields the removal of one or more gears from the transmission assembly **16**. Efficiency of a cutting performance of the band saw **10, 310** is thereby increased and leads to an exceptional run time of the band saw **10, 310**. Removal of one or more gears also improves manufacturability and eases the process of assembling the band saw **10, 310** by reducing cost of assembly and increasing speed of the assembling process.

[0041] Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described.

[0042] Various features of the present invention are set forth in the following claims.

Claims

1. A band saw comprising: a housing including a handle configured to be grasped by a user during a cutting operation; a motor supported by the housing; a drive wheel assembly rotationally driven by the motor, the drive wheel assembly including a drive wheel and a driven wheel; an opening defined within the housing and located between the drive wheel and the driven wheel; an endless saw blade driven by the drive wheel assembly and configured to pass through the opening to cut a workpiece during the cutting operation; and a transmission assembly having an input member for receiving torque from the motor and an output member for transmitting torque to the drive wheel about a drive wheel axis, wherein the transmission assembly includes at least one cycloidal gear stage.
2. The band saw of claim 1, wherein the at least one cycloidal gear stage of the transmission assembly includes a first cam coupled to the input member for rotation about a first axis, and a first disc defining a second axis and configured to both rotate about the second axis and orbit about the first axis, thereby undergoing cycloidal motion, in response to rotation of the first cam about the first axis.
3. The band saw of claim 2, wherein the transmission assembly further includes a transmission housing in which the first cam and the first disc are located, and a plurality of cylindrical pins affixed within the housing that are engaged by teeth on the first disc during cycloidal motion of the first disc.
4. The band saw of claim 3, wherein the transmission housing includes a frame and a cover coupled to the frame, and wherein the cylindrical pins extend between the frame and the cover.
5. The band saw of claim 3, wherein the drive wheel includes a plurality of drive lugs extending therefrom in a direction parallel to the drive wheel axis, wherein the first disc includes a plurality of apertures in which the drive lugs are received, respectively, and wherein an inner diameter of the apertures is greater than an outer diameter of the drive lugs to provide clearance for the first disc during the cycloidal motion thereof.
6. The band saw of claim 3, wherein the plurality of cylindrical pins includes N pins and the first disc includes N-1 teeth.
7. The band saw of claim 3, wherein the plurality of cylindrical pins includes 45 pins and the first disc includes 232 teeth to provide a gear ratio of the transmission assembly of 232:1.
8. The band saw of claim 2, wherein the transmission assembly further includes a vibration attenuating assembly configured to move out of phase with the first cam and the first disc.
9. The band saw of claim 8, wherein the vibration attenuating assembly includes a second cam coupled to the input member for rotation about the first axis, the second cam being 180 degrees out of phase with the first cam, and a second disc defining a third axis and configured to both rotate about the third axis and orbit about the first axis, thereby undergoing cycloidal motion, in response to rotation of the second cam about the first axis.

10. The band saw of claim 9, wherein the drive wheel includes a plurality of drive lugs extending therefrom in a direction parallel to the drive wheel axis, wherein the first disc includes a first plurality of apertures and the second disc includes a second plurality of apertures, wherein the drive lugs are received in both the first and second pluralities of apertures, and wherein an inner diameter of the first plurality of apertures and an inner diameter of the second plurality of apertures are greater than an outer diameter of the drive lugs to provide clearance for each of the first disc and the second disc during the cycloidal motion thereof.

11. The band saw of claim 1, wherein the housing includes a deck and a motor housing portion extending from the deck, and wherein a height dimension of the motor housing portion from the deck is 25.5 mm.

12. A band saw comprising: a housing including a handle configured to be grasped by a user during a cutting operation; a motor supported by the housing, the motor including an outer rotor, and an inner stator at least partially received within and surrounded by the outer rotor; a drive wheel assembly rotationally driven by the motor, the drive wheel assembly including a drive wheel and a driven wheel; an opening defined within the housing and located between the drive wheel and the driven wheel; an endless saw blade driven by the drive wheel assembly and configured to pass through the opening to cut a workpiece during the cutting operation; and a transmission assembly having an input member for receiving torque from the motor and an output member for transmitting torque to the drive wheel about a drive wheel axis, wherein the transmission assembly includes at least one cycloidal gear stage.

13. The band saw of claim 12, wherein the at least one cycloidal gear stage of the transmission assembly includes a first cam coupled to the input member for rotation about a first axis, and a first disc defining a second axis and configured to both rotate about the second axis and orbit about the first axis, thereby undergoing cycloidal motion, in response to rotation of the first cam about the first axis.

14. The band saw of claim 13, wherein the drive wheel includes a plurality of drive lugs extending therefrom in a direction parallel to the drive wheel axis, wherein the first disc includes a plurality of apertures in which the drive lugs are received, respectively, and wherein an inner diameter of the apertures is greater than an outer diameter of the drive lugs to provide clearance for the first disc during the cycloidal motion thereof.

15. The band saw of claim 13, wherein the transmission assembly further includes a vibration attenuating assembly configured to move out of phase with the first cam and the first disc.

16. The band saw of claim 15, wherein the vibration attenuating assembly includes a second cam coupled to the input member for rotation about the first axis, the second cam being 180 degrees out of phase with the first cam, and a second disc defining a third axis and configured to both rotate about the third axis and orbit about the first axis, thereby undergoing cycloidal motion, in response to rotation of the second cam about the first axis.

17. A band saw comprising: a housing including a handle configured to be grasped by a user during a cutting operation; a motor supported by the housing, the motor including an outer rotor, and an inner stator at least partially received within and surrounded by the outer rotor; a drive wheel assembly rotationally driven by the motor, the drive wheel assembly including a drive wheel and a driven wheel; an opening defined within the housing and located between the drive wheel and the driven wheel; and an endless saw blade driven by the drive wheel assembly and configured to pass through the opening to cut a workpiece during the cutting operation.

18. The band saw of claim 17, wherein the outer rotor has a rotor body and a plurality of permanent magnets supported on the rotor body, and wherein the inner stator has a stator mount, a stator core supported by the stator mount, and a plurality of coils, and wherein the rotor body surrounds the stator core, portions of the stator mount, and the plurality of coils.

19. The band saw of claim 17, wherein the outer rotor has a rotor shaft rotatably supported by at least one bearing such that the outer rotor rotates relative to the inner stator.

20. The band saw of claim 17, wherein the outer rotor includes a rotor frame having a plurality of blades and a plurality of apertures defined between adjacent blades, the plurality of blades configured to generate an airflow that passes through the apertures to cool the motor.
